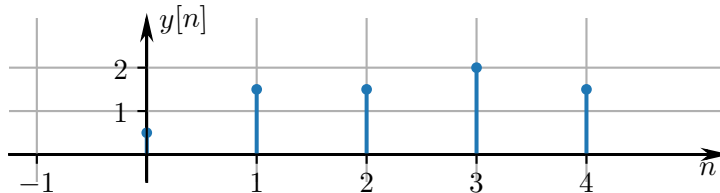


Q1. Consider the sequence $\{1, 2, 1, 3\}$, and answer the following questions:

- (a) Sketch the lollipop (stem) plot for this sequence.
- (b) Confirm the lollipop plot below if the sequence is filtered by the filter described by:

$$y[n] = \frac{1}{2} (x[n] + x[n-1]) . \quad (1)$$

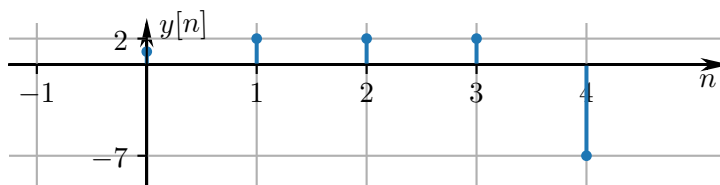


- (c) How long does this filter take to reach the steady-state response.
- (d) If Gaussian noise with independent samples is presented to the filter, by what factor is the noise standard deviation reduced by?
- (e) If a step change is applied to this filter, how many samples does the transient response last for?

Q2. Consider the sequence $\{1, 3, 5, 7\}$ and answer the following questions:

- (a) Sketch the lollipop plot for this sequence.
- (b) Confirm the lollipop plot below if the sequence is filtered by the filter described by:

$$y[n] = x[n] - x[n-1]. \quad (2)$$



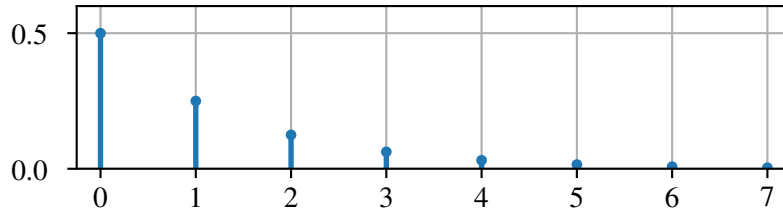
- (c) How long does this filter take to reach the steady-state response.
- (d) What does this filter do?

Q3. Consider the sequence $\{1, 0, 0, 0, 0, 0, 0, 0\}$ and answer the following questions:

- (a) Sketch the lollipop plot for this sequence.
- (b) Confirm the lollipop plot below if the sequence is filtered by the filter described by:

$$y[n] = 0.5y[n-1] + 0.5x[n]. \quad (3)$$

Assume $y[n] = 0$ for $n < 0$.



- (c) Why do you think this response is called the impulse response of the filter?

Q4. The unilateral Z-transform of a sequence $x[n]$ is defined by:

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n} \quad (4)$$

- (a) Determine the Z-transform of $ax[n] + by[n]$ using the definition of the unilateral Z-transform.
- (b) Determine the Z-transform of $x[n-1]$ using the definition of the unilateral Z-transform.
- (c) Determine the Z-transform of $0.7x[n] + 0.3x[n-1]$.
- (d) If $y[n] = 0.7y[n-1] + 0.3x[n]$ show that

$$Y(z) = \frac{0.3}{1 - 0.7z^{-1}}X(z).$$